

The Computable Continuum: Depth Without Breadth

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Abstract

The uncountability of the real numbers is widely regarded as a single mathematical fact, but it conflates two distinct properties: Archimedean completeness (depth) and set-theoretic cardinality explosion (breadth). We demonstrate that all physically relevant properties of the continuum — connectedness, Cauchy completeness with computable modulus, the intermediate value theorem for computable functions, and the ordered-field structure — are preserved by the computable real numbers $\mathbb{R}_c$, which is countable. We prove that two non-computable reals agreeing on all rational approximations with computable error bounds are physically indistinguishable: no finite measurement protocol can produce distinct predictions for them. The Continuum Hypothesis, Chaitin’s Omega, and the existence of non-computable reals are mathematically true (in ZFC) but physically vacuous. The operational ontology of the physical continuum is the real numbers $\mathbb{R}_c$, not the real numbers — depth without breadth.

1. Introduction: The Conflation Problem

1.1 Three Facts, Two Properties

Consider three mathematical facts about the real numbers $\mathbb{R}$:

1. **$\mathbb{Q}$ is dense but $\mathbb{R}$ is complete:** Every Cauchy sequence of rationals with a rational modulus of convergence has a limit in $\mathbb{R}$, but not in $\mathbb{Q}$. This is the **depth** property — the continuum is infinitely fine-grained, with no gaps.
2. **Most real numbers are non-computable:** Chaitin’s Ω , the Specker sequence limit, and continuum-many other reals cannot be approximated by any finite Turing machine with computable error bounds. This is the **breadth** property — the power set of $\mathbb{N}$ generates vastly more “points” than can be individually named.
3. **The Continuum Hypothesis:** There is no set S with $|\mathbb{N}| < |S| < |\mathbb{R}|$. Under CH, $|\mathbb{R}| = \aleph_1$. This is a statement about **breadth** — it concerns the cardinality of the power set, not the metric structure of the continuum.

Standard mathematical pedagogy treats these as consequences of a single fact: “ $\mathbb{R}$ is uncountable” (Cantor’s diagonal argument). But the diagonal argument proves breadth — it constructs a real number not in a given enumeration. The completeness of $\mathbb{R}$ (depth) is a separate axiom (Dedekind completeness or Cauchy completeness) with different ontological status.

This conflation creates a persistent confusion in the foundations of physics. Physicists correctly observe that physics requires limits, continuity, and connectedness — properties of the completed continuum. They incorrectly infer that physics therefore requires the full set-theoretic power of

$\mathbb{R}$, including its uncountable cardinality and the continuum hypothesis. The middle term does not imply the conclusion.

1.2 The Thesis

We demonstrate that:

1. **The computable real numbers** $\mathbb{R}_c$ form a real closed field, are connected (no computable nontrivial clopen sets), and are complete with respect to computable Cauchy sequences with computable moduli of convergence. They preserve every physically relevant property of $\mathbb{R}$ while being countable ($|\mathbb{R}_c| = \aleph_0$).
2. **Non-computable reals are physically unfalsifiable:** For any two non-computable reals x, y that agree on all rational approximations with computable error bounds, no finite measurement protocol can produce measurement-distinguishable predictions parameterized by x versus y .
3. **The Continuum Hypothesis has zero physical content:** It quantifies over set-theoretic objects (the power set of $\mathbb{N}$) that are not required for the physical description of the continuum.

The operational ontology is: $\mathbb{R}_c$, **not** $\mathbb{R}$, **is the physical continuum**. Depth stays; breadth goes.

2. Preliminaries

2.1 Computable Real Numbers

Definition 2.1 (Computable Real). A real number $x \in \mathbb{R}$ is *computable* if there exists a Turing machine T such that for all $n \in \mathbb{N}$, $T(n)$ halts and outputs a rational number q_n with $|x - q_n| < 2^{-n}$.

The function $n \mapsto 2^{-n}$ is the *modulus of convergence*. The sequence $\{q_n\}$ is a *computable Cauchy sequence with computable modulus*.

Definition 2.2 ($\mathbb{R}_c$). $\mathbb{R}_c = \{x \in \mathbb{R} : x \text{ is computable}\}$ is the set of *computable real numbers*.

Basic facts (proofs in standard computable analysis literature, e.g., Weihrauch 2000, Pour-El & Richards 1989): - $\mathbb{Q} \subset \mathbb{R}_c$, all algebraic numbers are in $\mathbb{R}_c$ - π, e, γ , and all standard mathematical constants are in $\mathbb{R}_c$ - $\mathbb{R}_c$ is closed under $+, \times, -, \div$ (by non-zero), and absolute value - $\mathbb{R}_c$ forms an ordered field - $|\mathbb{R}_c| = \aleph_0$ (since there are countably many Turing machines)

2.2 Completeness and Connectedness

Definition 2.3 (Computable Completeness). A metric space (X, d) is *computably complete* if every computable Cauchy sequence $\{x_n\} \subset X$ with computable modulus of convergence has a limit in X .

Theorem 2.4 ($\mathbb{R}_c$ is computably complete). Let $\{x_n\} \subset \mathbb{R}_c$ be a computable sequence with computable modulus $m: \mathbb{N} \rightarrow \mathbb{N}$ (i.e., $|x_i - x_j| < 2^{-k}$ whenever $i, j \geq m(k)$). Then $\lim x_n \in \mathbb{R}_c$.

Proof. The standard diagonal argument: construct a Turing machine that, on input k , computes $x_{m(k+1)}$ to precision $2^{-(k+1)}$ and outputs the resulting rational. By the triangle inequality, this approximates the limit to within 2^{-k} . Done

Theorem 2.5 ($\mathbb{R}_c$ is connected). There exists no computable nontrivial clopen subset of $\mathbb{R}_c$. Equivalently: no computable separation of $\mathbb{R}_c$ into two disjoint nonempty open sets.

Proof. If $U \subset \mathbb{R}_c$ were computable, clopen, and nontrivial, then the characteristic function χ_U would be computable. But a computable total function on $\mathbb{R}_c$ with values in $\{0, 1\}$ that is both open and closed would violate the intermediate value property for rational endpoints — a contradiction since $\mathbb{R}_c$ satisfies the IVT for computable functions (see Theorem 2.7). Done

Corollary 2.6. $\mathbb{R}_c$ has no “gaps” detectable by computable means. Any apparent gap would require a non-computable Dedekind cut to specify.

2.3 Intermediate Value and Real Closure

Theorem 2.7 (Intermediate Value for $\mathbb{R}_c$). Let $f: \mathbb{R}_c \rightarrow \mathbb{R}_c$ be a computable function. If $a < b$ are computable reals with $f(a) < 0 < f(b)$, then there exists a computable $c \in (a, b)$ with $f(c) = 0$.

Proof. Bisection is computable: at each step, evaluate f at the midpoint to precision sufficient to determine its sign. The sequence of midpoints is a computable Cauchy sequence with modulus — its limit exists in $\mathbb{R}_c$ by Theorem 2.4. Done

Theorem 2.8 ($\mathbb{R}_c$ is a real closed field). $\mathbb{R}_c$ is elementarily equivalent to $\mathbb{R}$ in the language of ordered fields. Every positive computable real has a computable square root; every odd-degree polynomial with computable coefficients has a computable root in $\mathbb{R}_c$.

Proof. The square-root algorithm (Newton’s method or digit-by-digit extraction) is computable. For odd-degree polynomials, Sturm’s theorem provides a computable root-isolation procedure; the limit of the isolating intervals gives a computable root. Done

3. The Specker Sequence: The Precise Boundary

3.1 Construction

The Specker sequence (Specker 1949) is the canonical example demarcating depth from breadth.

Let $\{\phi_e\}$ be a computable enumeration of partial recursive functions. Define:

$$a_n = \sum_{\substack{e \leq n \\ \phi_e(e) \text{ halts in } \leq n \text{ steps}}} 2^{-e}$$

Properties: - $\{a_n\}$ is a computable sequence of rational numbers PASS - $\{a_n\}$ is strictly increasing PASS - $\{a_n\}$ is bounded above (by 1) PASS - The limit $x^* = \lim a_n$ exists in $\mathbb{R}$ (by monotone convergence) PASS - x^* is *not* computable: its binary expansion encodes the halting problem FAIL - The sequence has *no computable modulus of convergence*: knowing how fast it converges would require solving the halting problem FAIL

3.2 Physical Interpretation

The Specker sequence is a bounded, computable, increasing sequence of rational numbers whose limit is physically inaccessible — it can be approximated arbitrarily well by rationals (each a_n is a rational), but no finite protocol can tell you *how close* you are to the limit at any finite stage.

This is the **precise boundary** between depth and breadth:

- **Depth = computable Cauchy sequences with computable modulus.** Every measurement protocol produces such a sequence: you measure, compute error bounds, iterate. The limit exists in $\mathbb{R}_c$.
- **Breadth = arbitrary Cauchy sequences (with no computable modulus).** Set-theoretically, these converge in $\mathbb{R}$. Operationally, they define numbers with no finite protocol for localization.

The Specker sequence has depth (its terms are computable) but not physical depth (its modulus of convergence is not). Its limit belongs to breadth — the dark continuum of non-computable reals.

4. Physical Unfalsifiability of Non-Computable Reals

4.1 Measurement Protocols as Computable Functionals

Definition 4.1 (Physical Measurement Protocol). A *physical measurement protocol* P is a finite specification of:

1. A preparation procedure (classical description of the initial state)
2. An interaction procedure (coupling to a measurement apparatus)
3. A readout procedure: on input rational $\epsilon > 0$, $P(\epsilon)$ halts and outputs a rational q with a guarantee that the “true” measurement outcome is within ϵ of q .

The finiteness condition means P is a Turing machine. The readout procedure defines a computable functional $\Phi_P: \mathbb{R} \rightarrow \mathbb{Q}$ that maps a real parameter to a rational approximation.

Definition 4.2 (Physical Distinguishability). Two real numbers $x, y \in \mathbb{R}$ are *physically distinguishable* if there exists a physical measurement protocol P such that $|\Phi_P(x) - \Phi_P(y)| > \epsilon_P$ for some fixed $\epsilon_P > 0$ — i.e., P produces measurement-distinguishable outputs when parameterized by x versus y .

4.2 Main Theorem

Theorem 4.3 (Unfalsifiability of Non-Computable Reals). Let $x, y \in \mathbb{R}$ be two non-computable real numbers that are Turing-equivalent (each computes the other relative to an oracle). If x and y agree on all rational approximations with computable error bounds — i.e., for every computable Cauchy sequence $\{q_n\}$ with computable modulus that converges to x , $\{q_n\}$ also converges to y — then x and y are physically indistinguishable.

Proof. Let P be any physical measurement protocol. By Definition 4.1, the readout functional Φ_P is computable. Since Φ_P is computable, its output $\Phi_P(x)$ depends only on a finite initial segment of a Cauchy sequence approximating x with computable modulus.

Specifically: there exists a computable function $m: \mathbb{N} \rightarrow \mathbb{N}$ such that if $q, r \in \mathbb{Q}$ satisfy $|x - q| < 2^{-m(k)}$ and $|x - r| < 2^{-m(k)}$, then $|\Phi_P(q) - \Phi_P(r)| < 2^{-k}$. The function m is the continuity modulus of Φ_P at x .

Given $\epsilon > 0$, choose k with $2^{-k} < \epsilon/2$. Let q_n be a computable Cauchy sequence with computable modulus that converges to x . Then for sufficiently large n , $|x - q_n| < 2^{-m(k)}$.

By hypothesis, $\{q_n\}$ also converges to y , so $|y - q_n| < 2^{-m(k)}$ for sufficiently large n .

Therefore $|\Phi_P(x) - \Phi_P(y)| \leq |\Phi_P(x) - \Phi_P(q_n)| + |\Phi_P(q_n) - \Phi_P(y)| < 2^{-k} + 2^{-k} < \epsilon$.

Since $\epsilon > 0$ was arbitrary, $\Phi_P(x) = \Phi_P(y)$. Done

Corollary 4.4 (Dark Continuum). There exist continuum-many non-computable reals that are pairwise physically indistinguishable. Each is physically equivalent to some computable real (its “computable shadow” — the closest computable approximation, which exists by the density of $\mathbb{R}_c$ in $\mathbb{R}$).

4.3 The Continuum Hypothesis Has Zero Physical Content

Theorem 4.5. The truth value of the Continuum Hypothesis (CH) cannot affect any measurement-distinguishable physical prediction.

Proof. CH is a statement about the cardinality of the power set of $\mathbb{N}$ — specifically, whether there exists a subset of $\mathbb{R}$ with cardinality strictly between $\aleph_0$ and $2^{\aleph_0}$. But Theorem 4.3 shows that non-computable reals are physically indistinguishable from each other and from their computable shadows. The excess cardinality contributed by non-computable reals cannot generate measurement-distinguishable predictions. Therefore, whether $|\mathbb{R}| = \aleph_1$ or $|\mathbb{R}| > \aleph_1$ cannot affect the output of any physical measurement protocol. Done

Remark. This does not mean CH is “meaningless” mathematically. It means CH quantifies over objects — specific non-computable subsets of $\mathbb{R}$ — that are physically inaccessible. The distinction between $\aleph_1$ and $\aleph_2$ is a distinction between unfalsifiable cardinalities. Physics is indifferent to it.

5. Operational Ontology

5.1 The OC Criterion, Formalized

Definition 5.1 (Physically Real Quantity). A real number x is *physically real* iff there exists a Turing machine T such that for all rational $\epsilon > 0$, $T(\epsilon)$ halts and outputs a rational q with $|x - q| < \epsilon$.

Equivalently: $x \in \mathbb{R}_c$ — x has a computable binary expansion (or any computable base expansion; computability is base-invariant).

This is the Ontological Closure (OC) criterion specialized to real numbers: a quantity has physical reality only if it is the limit of a finite D/R (description/realization) procedure with a computable modulus of convergence.

5.2 What This Excludes

The OC criterion excludes:

Object	Reason for Exclusion	Physical Cost
Non-computable reals (Chaitin's Ω , etc.)	No finite protocol produces them with computable error bound	None — they are unfalsifiable (Theorem 4.3)
The Continuum Hypothesis	Quantifies over physically inaccessible sets	None — cannot affect any measurement (Theorem 4.5)
Arbitrary Dedekind cuts	May require non-computable specification	None — the computable cuts are sufficient for all computable suprema
The full power set of $\mathbb{N}$	Generates physically equivalent reals	None — the excess reals are physical duplicates of computable ones

5.3 What This Preserves

The OC criterion preserves:

Object	Why It Survives	Physical Role
All rationals	Finite descriptions, computable	Approximate all physical quantities
All standard constants (π , e , γ , ...)	Computable via standard algorithms	Dynamics, geometry, statistics
Completeness under computable limits	Theorem 2.4	All measurement sequences converge
Connectedness	Theorem 2.5	Causality, time ordering, topology
Intermediate value	Theorem 2.7	Existence proofs for physical equilibria
Real closure	Theorem 2.8	Algebraic closure of physical quantities

6. Implications for Physics

6.1 The Continuum Is Smaller Than You Think

The physical continuum is $\mathbb{R}_c$, not $\mathbb{R}$. $\mathbb{R}_c$ is countable. But — crucially — it is not *computably* countable. There is no Turing machine that enumerates $\mathbb{R}_c$. So the fact that $|\mathbb{R}_c| = \aleph_0$ is a meta-theoretic observation with no operational content. Inside the physical theory, $\mathbb{R}_c$ behaves exactly like $\mathbb{R}$: it's connected, complete under computable limits, and satisfies all the first-order sentences of real-closed fields.

The “uncountability” of $\mathbb{R}$ is breadth — a set-theoretic artifact. The physically real content is depth — the Archimedean completeness that permits limits, continuity, and dynamics. By conflating these two, standard mathematical physics has imported a vast structure (the power set of $\mathbb{N}$) that has no physical consequences and whose entities (non-computable reals) are pairwise physically indistinguishable.

6.2 Renormalization and Infinities

The standard narrative is that renormalization “cures” the infinities of quantum field theory by absorbing them into finite parameters. From the OC perspective, the infinities never existed physically — they are artifacts of integrating over the full real line $\mathbb{R}$ instead of the physically relevant $\mathbb{R}_c$.

A computable integral $\int_a^b f(x) dx$ exists in $\mathbb{R}_c$ whenever f is computable and the integral converges with computable modulus. The “ultraviolet divergences” of QFT correspond to integrals that fail this condition — they are not computable, hence not physically real. Renormalization is not “subtracting infinities” but recognizing that the naïve continuum integral over $\mathbb{R}$ generated a quantity with no physical reality, and the correct quantity is the computable integral over $\mathbb{R}_c$.

6.3 Determinism and Chaos

The existence of non-computable reals is sometimes invoked to explain why deterministic systems can exhibit apparent randomness (chaotic dynamics amplify uncomputable initial conditions). But this argument requires that the initial conditions *could have been* non-computable. Under OC, they cannot be — all physically real initial conditions are computable. The apparent randomness of chaotic systems must arise from another source (sensitivity to finite-precision measurement, not to genuinely uncomputable structure).

7. Relation to the Trilogy

This paper establishes that:

1. The physical continuum is $\mathbb{R}_c$, not $\mathbb{R}$ — **depth without breadth**.
2. The uncountability of $\mathbb{R}$ (breadth) is physically vacuous — non-computable reals are unfalsifiable, and the Continuum Hypothesis has zero physical content.
3. The OC criterion applied to real numbers yields $\mathbb{R}_c$ — which is countable but connected, complete under computable limits, and sufficient for all physical measurement.

Paper II (P-adic Spin, Information, and Ultrametric Internal Quantum Numbers) develops the second thread: the p-adic completions of $\mathbb{Q}$ are physically real not as geometric dimensions but as valuation-theoretic dimensions encoding spin, internal quantum numbers, and information-theoretic structure.

Paper III (A Unified Ontology of the Physical Continuum) synthesizes both threads into a unified adelic framework and completes the argument that the three axes — depth, breadth, and valuation — partition the completions of $\mathbb{Q}$ by physical status.

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